

# Self-Application Is Not Free: the Monitoring Channel Is Part of the Failure Channel, and the Self Does Not Survive Its Own Reconstruction

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**Status.** Unified paper (2026-10-03). Two linked claims, one measured honestly at full size and one measured honestly at null. Everything substantive is marked **MEASURED** / **INTERPRETATION** / **PROJECTION** (§0). The work-score consequence of the mechanism is a **NULL** — directional, unstable across two independent sets, never significant at any n run (§10) — and this paper does not soften that. What is established is the mechanism itself, replicated across two independent sets and two pins to within ~0.4% on its headline quantity (§9, §10), and the honest negative beside it.

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## 0. Marking convention, provenance, and the unit

Every substantive claim is marked in place:

| mark                  | meaning                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------|
| <b>MEASURED</b>       | a number or outcome produced by a committed driver or a real run, with its artifact named |
| <b>INTERPRETATION</b> | a reading of measurements in other vocabulary                                             |
| <b>PROJECTION</b>     | a prediction or extrapolation with no measurement behind it yet                           |

**Provenance series.** [R-n] names a measurement in the methods-half results record (paper2/paper-2-results-record.md): R1 the monitoring carrier, R2 the instrument validation, R3 the saturation, R4 the within-window negative, R5 the prompt-coupling (crutch) finding, R6 the render-order void. The deciding set’s own measurements (this paper’s results half) cite their artifact paths inline (runs/p3-set, runs/p3-set2). Paper 1 (Marshall, *Self-Application as the Common Source of Benefit and Failure*) is **cited, not re-derived**: its §4.9 (the border-collision fold), §5.3 (the D/G dissociation), §4.5 (the standing-cost budget) carry the model-side ground of every model claim made here.

**The honest unit, stated once, up front.** This paper reports two experiments on two different units, and neither is “an agent” in general. The **methods half** (§3–§6) measured a **self-referential substrate inside a measurement rig** — a turn loop whose self-block thins, wired to a frozen plant — with no task, no execution, no entailment and no scheduled consolidation in the build; the rig is not a worker, and paper 2’s own §1.1 said so at equal length. The **results half** (§7–§10) measured a **worker**: one loop, one 12-target task suite, real tools, an external evaluator, one model (qwen3.5:9b). Claims about “agents” in general are marked PROJECTION and are not made as findings.

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## Abstract

Dual control — the closest control-theoretic precedent for the cost of monitoring — assumes the sensing channel is orthogonal to the controlled process: the probe is chosen, bounded and instrumental, and monitoring costs performance rather than damaging the plant. Paper 1 established, in a five-state dynamical model, that a self-referential control loop breaks this assumption structurally: monitoring *is* self-reference; its cost **feeds the failure variable**; the sensor is **destroyed by the failure it must detect** (at the collapsed fixed point  $D = 0.5455$ , exactly its baseline, while  $G$  collapses  $0.8855 \rightarrow 0.0485$ ); the collapse transition is a **border-collision fold** — discontinuous by construction, no intermediate state; and the standing cost of self-application has a sharp measured budget ( $c_{\text{int}} \approx 0.112$ ). This paper takes those predictions to two real substrates, in two stages, and reports one replicated mechanism, one instrument finding **that is the same failure as the mechanism, measured in the paper’s own apparatus** (the monitoring channel and the measurement channel share the failure of the thing they measure), and one honest null.

**Stage one — the rig.** A self-referential substrate (a turn loop whose self is a four-step derivation, priced per turn, injected per window) inside a measurement rig. MEASURED: the monitoring paragraph’s mere presence costs the process it monitors — **+329 chars/turn of context, moving the compaction interval +0.625 turns, permutation  $p = 0.0016$** , with 84% of the effect being the paragraph’s own char cost [R1]. Also measured, and it is the paper’s claim occurring in its own apparatus: **every self-side instrument failed because the instrument and the thing measured share a channel** — the harness describes the self to the agent in every prompt, and removing that description drops the reconstruction-fidelity reading from 0.75 to 0.00 while the agent writes *more* text [R5]; the instrument saturates where the signal dies [R3]; and the one behavioural DV tried was decided by render order 135/135 and is void [R6]. The sensor-placement rule this forced — *put the sensor outside the channel it measures* — is the design rule the worker was built to.

**Stage two — the worker.** Three arms, one loop, one 12-target task suite, real tools, a sealed held-out evaluator outside the agent’s write domain, the plant edge severed in every arm; 60 turns; two independent sets ( $n = 8/\text{arm}$  at pin 645b923,  $n = 10/\text{arm}$  at pin bfbe5d0). MEASURED, and replicated to within 0.4% on the headline quantity: **an agent that must reconstruct its own self after every compaction loses the self in every cell, permanently — empty-self fraction 1.000 in 8/8 then 10/10 D1 cells, against 0.08/0.07 in the inject control, where every gap is a single turn at a compaction boundary — while spending a median ~400,000 chars of self-directed thinking per cell (~117,000–150,000 eval tokens) on exactly that reconstruction, at a thinking-to-output ratio of ~21×**. The loss is categorical, not graded: `self_steps` takes only [] or [1,2,3,4], never a partial value, in any of the 3,123 rows of the two sets — which is precisely the transition type paper 1’s border-collision fold predicts (present-in-full or absent-entirely, zero intermediates). The pre-registration’s co-movement requirement is **withdrawn as mis-specified**: a fold has no gradient to co-move with.

**The work consequence is a NULL, reported as one.** The pre-registered secondary DV orders as the claim predicts — the arm that pays for self-maintenance is worst in both sets and on both

DVs — but the effect **halved on replication** (D0 – D1: 2.26  $\rightarrow$  1.08 on the 0–12 ordinal;  $p \approx 0.10 \rightarrow 0.35$ ), the carrier contrast is null in set 2 (ordinal  $p = 0.96$ ; finer-DV  $p = 1.00$ ), and the finer continuous DV bought for the re-run **failed to cut the variance it was bought to cut** (within-arm sd essentially unchanged) — a negative instrument result. An effect that halves on replication and reaches  $p < 0.05$  in neither set is **not established**, and nothing in this paper claims the agent’s work was damaged. What is established: **self-application is not free** — the monitoring channel consumes the process’s own resources by its presence, and the self, once it must be reconstructed rather than restored, does not survive the reconstruction — at a cost of order  $10^5$  tokens per run, in a working agent, on a real task, with the plant unable to explain any of it.

**Keywords:** self-application; self-reference; dual control; monitoring cost; sensor placement; long-horizon agents; reconstruction; depersonalization-derealization (DPDR)

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## 1. Introduction

Paper 1 (*Self-Application as the Common Source of Benefit and Failure*) established, in a deterministic five-state model, that self-application is the common source of benefit and failure: the same self-referential operation that gives the system its benefit also drives its collapse, and the collapse is a discontinuous annihilation on a switching manifold, not a gradual erosion. The model’s agent-level sentences were marked interpretations — the mapping from model slots to any real agent is ordinal and unvalidated. This paper is the test on real substrates, in two stages, and its title is its thesis in two claims:

1. **The monitoring channel is not orthogonal to the controlled process — it is (part of) the failure channel.** Monitoring is not an instrument the system can pick up and put down; it is the system’s own operation turned on itself, and its cost lands on the same resources the work runs on.
2. **In this working agent, the self does not survive its own reconstruction.** When the self must be rebuilt by the agent’s own generation after each memory compaction — rather than restored by the harness — the self is lost, categorically and permanently, at enormous generation cost, in every cell.

The two claims are **one structure seen twice — but across two units, not in one shared measurement**. No single experiment measures both. Claim (i) — the monitoring channel is (part of) the failure channel — is established on the **rig** (R1: +329 chars/turn, +0.625 turns,  $p = 0.0016$ ) and in the **model** (the D/G dissociation, §3); claim (ii) — the self does not survive reconstruction — is established on the **worker** (18/18 cells, §8). The worker-side contrast that would *independently* show claim (i) on the worker — the carrier, D1’ vs D0 — is **null** in set 2 and unresolved in set 1 (§10), so claim (i)’s worker-side evidence is the **cost** (R1 generalized: the ~400k chars of self-directed thinking per cell, the ~21 $\times$  ratio, the self-maintenance tokens against zero in both controls), not the failure-channel *identity*. What joins the two claims is **not a common experiment** but (a) the model’s **prior prediction** — the worker’s transition type (a discontinuous, all-or-nothing loss) was predicted in advance (§3, §9) and matched — and (b) R5, where claim (i)’s structure **recurs in the paper’s own instrumentation**: the instrument built to measure the self was supplied the self in every prompt, so it measured the rig — the sensor sharing the channel with its signal loses

the signal exactly when it is needed, the same structural fact as the D/G dissociation, measured in the apparatus built to test it. The “one structure seen twice” is a **structural correspondence across units**, not a shared measurement, and a reader should finish this section knowing exactly which claim rests on which unit.

**The arc, and the unit, honestly named.** §3–§6 are the methods half: the rig, its campaign, and findings R1–R6 — one positive cost result and five instrument findings, of which R5 is the most important (it is this paper’s own claim occurring in its own apparatus). §7–§10 are the results half: the worker’s deciding set, the mechanism, the fold correspondence, the honest null (with its replication). The rig is not a worker; the worker is one model on one task suite; and neither licenses a claim about agents in general. What the two halves share is the structure the model predicted, and the discipline of stating each result at exactly its measured size.

## 2. The broken assumption, and the rule that follows

Dual control (Feldbaum 1960–61, 1965) fully contains the tradeoff everyone reaches for first — probing versus directing, the Bar-Shalom–Tse dual effect, even an information-channel feedback loop (cautious-control turn-off). What it does not contain is the case where the act of monitoring worsens the monitored state, or where the sensor is destroyed by the failure it exists to detect:

| dual control assumes                                  | the self-referential case                                                                                                  |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| the probe is chosen and bounded                       | monitoring <b>is</b> self-reference — it cannot be made instrumental; it is the same operation the system already performs |
| the cost is performance (exploration vs exploitation) | the cost <b>feeds the failure variable</b> — monitoring adds inward attention, which is the collapse driver                |
| the sensor reads the plant from outside               | the <b>sensor is destroyed by the failure it must detect</b>                                                               |
| the fix is to probe less, or better                   | the fix is to <b>relocate the sensor</b>                                                                                   |

The break is structural, not a matter of degree. Under dual control “monitor less” is always available because the probe is instrumental; under the self-referential structure the monitoring load and the failure load are the same term in the same equation, so reducing monitoring reduces the benefit the system exists to have, while relocating the sensor changes which subsystem carries the signal when the failure arrives. The rule, which paper 1 measured the ground of but declined to state: **put the sensor in the subsystem that survives the failure, or outside the system entirely.** (Novelty caveat, carried forward verbatim from the methods half: sensor placement and observability are mature topics and a focused search of that literature is outstanding; “could not find it stated elsewhere” means not-yet-contradicted. **INTERPRETATION.**)

**The rule’s second life, discovered in this programme.** The rule was written for *protection*. The rig then showed it is also the rule for *measurement*: an instrument that routes through the

channel it measures (here: any self-side probe whose answer is supplied in the prompt) loses the signal exactly when it needs it [R5]. §7 states that finding; the worker’s evaluator was built to it.

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### 3. The model ground — cited from paper 1, not re-derived

Everything in this section is paper 1’s measurement, carried here as premise. This paper claims no credit for it and restates only what its own experiments lean on.

**The structure.** A five-state deterministic control system (a, G, D, S, g) whose failure mode is collapse of the self-content generator G into a self-maintaining state with no outward pull. Two couplings carry this paper: the **cannibalization channel** (the reducer reads the generator — D’s dynamics are driven by the switch state through  $B = \text{EMA}_{\tau} D(c)$ , so a reducer that must reduce the generator’s output raises demand for reduction; this is what makes the monitoring channel and the failure channel the same channel) and the **monitoring ceiling** ( $c_{\text{mon\_crit}} = 0.511$  gated / 0.112 ungated: above it, monitoring *sustains* the collapse it is supposed to detect).

**The D/G dissociation — “the sensor is destroyed,” measured.** At the collapsed fixed point, **D = 0.5455 — numerically its baseline, untouched — while G collapses 0.8855 → 0.0485** (paper 1 §5.3). A regulator whose sensor lives in G loses its sensor precisely when it is needed: the sensing channel is not merely perturbed by the failure but converted into the failure variable. **MEASURED** in-model.

**The standing-cost budget.** The window module’s collapse horizon is bounded by the normalized standing cost  $c_{\text{int}} = c_{\text{cap}} \cdot T_{\text{max}} / \tau_{\text{sim}} \approx 0.11223$  — identical across a  $4\times$  range of  $c_{\text{cap}}$ , and coincident with the frozen chronic-hold critical ( $a_{\text{hold\_crit}} = 0.11213$ ) and the ungated monitoring critical (0.112). Paper 1 states the agreement as a **consistency check, not an empirical unification** (the window’s cost channel was designed to enter where  $a_{\text{hold}}$  enters); the operative content here is the existence of a **sharp standing-cost budget** on self-application. Under sustained (non-episodic) stress the sign reverses: more capacity *lowers* the sustained-drive threshold, because the standing cost dominates — the self is not free even when it is helping. **MEASURED** in-model.

**The border-collision fold — the transition type this paper’s worker test leans on.** Paper 1 §4.9: on the chronic inward-drive axis, the healthy equilibrium is annihilated **exactly on the cannibalization switching manifold  $E = \Theta_{\text{eff}}$**  ( $\epsilon_c = 0.265192279$ ; residual  $|E - \Theta_{\text{eff}}| = 1e-13$  at the continuation’s last resolvable point; the tanh switch at `dpdr/model.py:93`). At the fold **no eigenvalue reaches zero** (healthy spectrum  $[-0.0015, -0.0055, -0.0088, -0.042702, -0.5978]$ ) — this is a border-collision fold, not a saddle-node — and there is **no critical slowing down** at any proximity (the dominant eigenvalue is bit-identical at  $-0.0015$  across five decades of distance). Across the swept  $\alpha G \times \text{duration}$  grid, **214 healthy / 98 stuck cells and zero intermediate outcomes** (§4.1). The model’s real early-warning signal is **basin retreat**, not slowing. **The prediction that matters here: the transition is present-in-full or absent-entirely — discontinuous by construction, with no intermediate state to find.**

**A negative that belongs in the record.** The dual-control-friendly reading — “the coupling’s raised rescue cost ( $\sim 2.7\times$ ) is selective, buying durability” — was pre-registered against and **falsi-**

**fied** in-model (exp25: zero relapses in either arm, equal dwell at the arms' own thresholds, the prevented set's minimum OFF dwell 110.4 t.u. against a 50 t.u. fragility cutoff). The raised dose is pure cost. The mirror reading — “monitor more carefully” as a fix — is thereby also without support. **MEASURED.**

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## 4. Related work

**This section is positioned here, not at the front, because both claims are corrections to specific precedents, and a reader needs the model's own numbers (§3) to weigh what each precedent does and does not cover.** Each entry is related to the claim it bears on and how — agreeing, complicating, or not engaging — rather than listed. Read-status is stated where it is less than full, exactly as paper 1 states it; nothing here is invented.

### 4.1 Dual control — the closest precedent, and the assumption this paper breaks

Dual control (Feldbaum 1960–61, 1965) is the control-theoretic precedent for the cost of monitoring, and paper 1 read it through the project's dual-control record (two modern surveys end-to-end: Mesbah 2018; Meijer & Rantzer 2026, arXiv 2608.20073). It fully contains the tradeoff everyone reaches for first — probing versus directing, the Bar-Shalom–Tse dual effect, cautious-control turn-off — and even a feedback loop through the *information state* (turn-off: low input → low information → high uncertainty → lower input). What it does **not** contain is the case this paper tests. The sensing equation carries noise only; the probe is *chosen, bounded and instrumental*; monitoring costs *performance*, it does not damage the plant; and the pathologies run in the opposite direction — they are failures of *too little* probing (turn-off, bursting), cured by more of it. No result in the surveys has the act of observing drive the observed system into a worse state, or a sensor destroyed by the failure it must detect (paper 1's dual-control audit, verdict (b) on claim 1: adjacent/variant — the cost half is classical, the failure-channel half is absent). **Bears on claim (i): complicates it.** This paper's contribution against this precedent is not “monitoring has a cost” — that is dual control's — but that in a self-referential system the monitoring load and the failure load are the same term in the same equation, so the cost is *pathogenic*, not a performance tradeoff, and the fix is *relocation*, not *reduction* (§2). The monitoring carrier this paper measures (R1: +329 chars/turn, +0.625 turns,  $p = 0.0016$ ) is a *cost* result of the kind dual control covers; the D/G dissociation (D untouched while G collapses) and the worker's self-loss are the *failure-channel* results it does not.

**Cited at second hand, stated as paper 1 states it:** Feldbaum's *Optimal Control Systems* (1965) was not read in the primary — it is available only under controlled digital lending (archive.org, access-restricted; OCR closed); the engagement here stands on the Meijer & Rantzer review read in full and the Filatov & Unbehauen textbook formulation, and has not been checked against Feldbaum's own statements (paper 1 ref [4], access level carried verbatim).

### 4.2 Self-reference, strange loops, and the classical self-application lineage

The structure this paper tests — a system whose operation turns on itself — is Hofstadter's strange loop (paper 1 ref [1], *Gödel, Escher, Bach*); the escape distinction is the philosophical vicious-vs-non-vicious circularity divide; and the formal content — which self-applications terminate — is

term-rewriting termination and confluence, undecidable in general (paper 1 refs [2], [3]). Paper 1’s window module derived its benefit term structurally from the self-improvement operator  $f = M \circ V \circ f_T$  of Zhang, Yuan & Zhang (2026, arXiv 2607.04277 — bounded self-simulation, evaluation, modification, with the fixed point from Kleene’s second recursion theorem; paper 1 ref [8]), whose bounded-horizon  $T$  is the shared axis of paper 1’s window experiment. **Bears on both claims: the framing.** This paper claims no discovery of the lineage; what it does is quantify, on two real substrates, one instance of the structure those works establish formally — and show (the worker) that the self’s bounded reconstruction fails at the boundary, which is a *dynamical* fact the lineage does not supply (Zhang supplies the operator anatomy, not a mechanism for degradation; the project’s literature search, JOB A, verified against the v2 full text).

**Cited at second hand:** paper 1 refs [2] (productive circularity) and [3] (term-rewriting termination/confluence) are marked ♦ in paper 1 — the specific canonical references were not pinned in the project record and the primaries have not been read; the class attributions stand on paper 1’s record.

#### 4.3 Criticality as a setpoint — and the “monitoring is free” assumption this paper contradicts

Hengen et al. (2025, paper 1 ref [6]) ask whether criticality is a unified setpoint of brain function — an optimum band, not a maximum, around which healthy brains tune themselves; Chialvo et al. (2020, ref [7]) implement self-tuning to a critical point via temporal correlations (Ising, Vicsek flocking, neural). **Bears on claim (i): the precedent this paper’s measured ceiling contradicts.** Paper 1 §5.5 notes that these window regulators *treat the monitoring channel as free* — the aggregate observable the controller reads costs nothing to compute; this paper’s measured monitoring carrier (R1) and the model’s monitoring ceiling ( $c_{\text{mon\_crit}} = 0.112$  ungated) are the same kind of constraint, but with the sensing channel *priced*: a self-directed monitoring load that consumes the process’s own resources is not free, and above the ceiling it sustains the failure it monitors. What transfers from this literature is the *setpoint concept* (a band, not a maximum) and the *budget*; what does not transfer is near-critical dynamics. Paper 1 §4.9 established the transition is a border-collision fold with no critical slowing down at any proximity (dominant eigenvalue bit-identical at  $-0.0015$  across five decades) — so the model is *not* an instance of the near-critical dynamics this literature describes, and its transition would not be preceded by the slowing signatures that literature predicts. The analogy is to the budget, not the dynamics (paper 1 §7.3, read in full).

**Cited at second hand:** Hengen et al. (ref [6]) is marked ♦ — article-level read not achieved (publisher blocked automated access; abstract verified via PubMed; paper 1’s access-level note carried verbatim). Chialvo et al. (ref [7]) was read in full via the arXiv preprint.

#### 4.4 Monitoring cost and the value of information in control

Dual control (§4.1) is the formal home of “information gathering has a cost traded against control.” Two adjacent branches bear on the placement of that cost. (a) **Partially observed stochastic control / measurable controls** (Runggaldier & Stettner 1994 and successors, surveyed by Yüksel) treats observation itself as a decision with a *resource* cost — pay to observe, budget to observe

— but finds only resource costs of observing, not state-degradation caused by observing (paper 1’s dual-control audit, §2). (b) **Adaptive- control fragility** (Rohrs, Valavani, Athans & Stein 1985; Anderson 1985 “bursting”) establishes that the adaptive machinery itself can destabilize the plant under unmodeled dynamics — but the destabilizer is the *adaptation/update*, triggered by external signals, not the sensing channel, and there is no sensor-death mechanism. **Bears on claim (i): complicates.** Both are kin to the relocation-vs-reduction distinction (§2): the standard move in fault detection and analytical redundancy is redundancy in an independent channel, which is precisely *relocation* — but neither branch states the failure-channel identity, the ceiling, or the sensor-death result this paper tests. A focused search of the sensor-placement and observability literature is **outstanding** (carried forward from paper 2’s §10); until it is done, “could not find it stated elsewhere” means not-yet-contradicted, not established novelty.

#### 4.5 Long-horizon LLM agents and context/memory management — where the worker sits

The worker’s results-half claim — that a working agent loses its self at the reconstruction boundary and pays  $\sim 21\times$  more generation than the self it recovers — sits in a literature that attributes long-horizon breakdown largely to *memory and context limits*. Bai et al., “The Long-Horizon Task Mirage? Diagnosing Where and Why Agentic Systems Break” (arXiv 2604.11978; paper 1 ref [12]), read in full by the project, gives the HORIZON benchmark, 3,100+ failure trajectories, a 7-category taxonomy; breakdown is “a structural shift in failure composition as horizon grows,” with planning errors, catastrophic forgetting, and memory limitations (context overflow, loss of earlier constraints during summarization) the dominant bottlenecks. **Bears on claim (ii): complicates.** That literature’s attribution is to the *container* (the context window, the summarizer); this paper’s worker result is a *control-structural* reframe of the same in-vivo failure — the self, carried as a derivation that must be reconstructed after each compaction, is annihilated discontinuously at the boundary, at a cost the literature measures as “memory limitation” but this paper reads as the monitoring channel consuming the process’s own resources. The two are not in conflict; this paper’s claim is about *which thing in the container* fails and *why categorically*, where the literature records *that* it fails.

Two further entries from the project’s literature search (JOB B6, abstract-level) sharpen the placement. Zerhoubi, Mitrovic & Granitzer (2026, “The Compaction Cliff in Long-Running AI Agent Memory,” arXiv 2608.22752) quantify in vivo that summarization preserves 53% of safety rules after one compaction round and 10% after five — the same compaction-destroys-content observation, not framed as a monitoring-channel identity. ACON (Kang et al. 2025, arXiv 2510.00615) shows content-aware compaction preserves  $>95\%$  task performance — an honest foil: compaction per se is not the claim; the *selector* and the *self-reconstruction* are. **Bears on claim (ii): the worker’s in-vivo observation has published company; none of it frames the loss as a control-theoretic fold.** The self-rewarding / judge-corruption line (Yuan et al. 2024, arXiv 2401.10020; Zhou 2026, arXiv 2607.05904 — both abstract-level via the search) instantiates “the evaluator is not orthogonal to what it evaluates” — a CLASS 2 cousin of R5, cited in the methods half’s positioning.

**Cited at second hand:** Zerhoubi et al., ACON, Yuan et al. and Zhou are abstract-level only (the search read abstracts, not full texts); Bai et al. (ref [12]) and the self-reflection entry (ref [13], Renze

& Guven, arXiv 2405.06682, read in full by the project) are the agent-literature primaries read in full. The Chen, Wang & Qu (2026) survey of recursive self-improvement (arXiv 2607.07663, 1,250 papers; paper 1 ref [21], read in full by the search) maps the field this paper sits in — its recurring diagnosis is evaluator-centric (“one bottleneck recurs everywhere: the evaluator”), and this paper quantifies one control-theoretic instance of that bottleneck.

#### 4.6 The DPDR / depersonalization literature — the phenomenon the model came from

The model’s origin is the depersonalization-derealization phenomenology, engaged exactly as paper 1 engages it (no clinical claim; the epistemology of the origin is paper 1’s). Deane, Miller & Wilkinson (2020, paper 1 ref [10], read in full) frame depersonalization as a loss of allostatic control in an active-inference system that “ceases to posit itself as a causally efficacious controller,” with a lock-in cascade and a predicted collapse of phenomenal depth — overlapping paper 1’s allostatic setpoint *S*, its loss of actuator authority, and its capture loop. Two structural differences paper 1 records: their cascade is ascending hierarchical, the model’s is single-level positive feedback; and their account is qualitative/architectural, the model supplies measured thresholds. Stephan et al. (2016, ref [5], read in full) propose “allostatic self-efficacy” — a metacognitive layer monitoring the system’s capacity to regulate its own states — with one architecture (their Fig. 7) that is the influencing-monitor structure paper 1’s *c\_mon* ceiling quantifies; but Stephan states no cost model and no ceiling, so the anticipation is architectural, not quantitative. Seth, Suzuki & Critchley (2012, ref [11], read in full) locate presence in interoceptive predictive coding, with the gain construct paralleling the model’s *g*. **Bears on claim (i): agreeing, at the structural level.** The D/G dissociation and the sensor-death result are a control-theoretic (non-Bayesian) formalization with measured thresholds of the allostatic-collapse structure these works describe qualitatively; this paper’s worker result is a substrate test of it, not a clinical claim.

The temporal-depth line (Tolchinsky et al. 2025, ref [16]; Madden & Serper 2026, ref [17]) proposes temporal-depth collapse as the mechanism of dissociation and shows temporal blurring/reduced future-self vividness predict depersonalization severity via perceived entrapment in a 433-person sample; the model’s *S* maps to the temporal-depth construct at construct level only, and entrapment has no state variable (paper 1 §7.6). The triple-network dissociation literature (Wei et al. 2024, ref [15], read in full; Carbone et al. 2025) supplies the model’s DMN/CEN/SN mutual-inhibition-under-SN-regulation characterization.

A note on the coercive-induction literature. The project’s cult-induction record (paper2/cult-induction-lit.md) maps Schein’s three stages (unfreezing/changing/refreezing) and Lifton’s eight criteria onto the model’s fold-and-holder structure, reading the cult as “the unheld crossing, engineered, with a malignant holder standing where the traditions put a benign one.” That mapping is an **INTERPRETATION** of a class of techniques, not a result of this paper, and is carried here only to name the boundary paper 1’s Non-use section draws: the thought-reform literature was converted into a warrant for extra-legal “deprogramming” (Young 2012, paper 1 ref [24], read in full), and this paper, like paper 1, makes no individual-level claim and licenses no intervention against a person’s stated wishes. **Bears on neither claim as evidence; stated so it is not silently absent.**

**Cited at second hand:** the clinical-instrument corpus (paper 1 ref [18]: Melges, Mathew, D’Souza, Colizzi, Simeon, Baker, Medford, Michal, Hunter, Pons, Leavitt, Guralnik, Sierra, Millman, Hammond) is cited via paper 1’s deposit (dpdr/external-validation.md) §8; the individual primaries are not read here and are carried at the access level paper 1 records. The cult-induction sources (Schein, Lifton, Singer, Stein, Hassan, Feliciano, Doychak, Bailey, Hadding, Pons) are framework/clinical/theoretical, not experimental, and are marked as such in cult-induction-lit.md; the mapping onto the model is an interpretation, stated there as inference, not measurement.

#### 4.7 What is load-bearing and what is not

The dual-control concession (§4.1) and the criticality-setpoint “monitoring is free” reading (§4.3) are load-bearing for claim (i): they are the precedents this paper’s measured ceiling and failure-channel identity are *against*, and the honesty of the contribution depends on naming them. The long-horizon-agent and compaction literatures (§4.5) are load-bearing for claim (ii): they are where the worker’s in-vivo observation has published company, and the contribution is the *control-structural reframe* of a failure those literatures record but do not theorize this way. The DPDR and self-reference lineages (§4.2, §4.6) supply the framing and the model’s origin; they agree at the structural level and this paper does not claim they anticipate its measured results. The outstanding search is sensor placement and observability (§4.4) — mature control-theory topics this paper is a substrate-sharing special case of, not a new problem class, and a focused search of that literature remains open.

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### 5. The self-referential substrate (the rig)

**Dropped from the unified paper, explicitly** (each maps to sections of the methods draft; the drop reason is one line each):

- **§4.1–§4.8 piece-by-piece build description** (the derivation-self, the priced reconstruction, the a\_hold seat, the commitment channel, the routing selector, the out-of-process boundary, the coverage mirror, the stated gaps). *Dropped:* the build detail is the methods-half draft’s content and its artifacts live in that deposit; what the unified paper needs from each piece is retained below at the size the worker test actually used it. The four unit-defining gaps (no task, no execution, no entailment, no scheduled consolidation) carry forward as the definition of “rig, not worker.”
- **§5.1–§5.2, §6.1, §6.4 — the stopped campaign’s six-condition matrix, its void and acceptance machinery.** *Dropped:* the campaign was stopped before its result was written up, superseded by the A/B redesign, and its prompt was measured backwards on both axes (below); a matrix that never produced its DV is not evidence in either direction.
- **§5.4–§5.5, §6.3 — the pilot’s five gates and the C2/C3 contrast.** *Dropped:* the pilot’s plant-side numbers stand in the record but its self-side attribution was withdrawn (R5), and the worker supersedes the plant-coupling question entirely by severing the plant edge — the worker’s arms are the design the pilot’s contrast wanted to be.
- **§7.3–§7.4 — the size theory and the predictor test.** *Dropped:* the predictor’s axis (trained-vs-templated) failed its own source cases and is recorded as future work (future-work-axis-

hunt.md); the precondition itself (the sign is per-model) is kept in §11's limits, and the size theory was never tested by anything this paper reports.

- **§8.3–§8.4's substrate-side verdicts (relocation-vs-reduction has no arm; the selector corruption arm) and §2.3's discriminating-test table.** *Dropped as verdicts:* both remain PROJECTION — no arm of any run in this paper tests them; the rule itself is carried (§2) with its status stated.
- **§9.2 corrections ledger, Appendices A/B.** *Dropped:* the methods draft carries them at full length and this paper cites that record rather than duplicating it; the corrections that bear on this paper's own design (the prompt confound, the plant-injection licence, the plant-edge severance) are kept in §5 below where they changed what was built. (Paper 2's §10 related work is **not** dropped — it is carried as §4 of this paper, expanded to stand alone, since the methods draft is held and unpublished.)

**What the rig was.** A turn loop that emits free-form text through a real model, extracts bracketed spans by a declared grammar, has them checked by a checker that resolves them by fact lookup, accrues the unchecked remainder as a priced backlog, lets its **self-block thin** under a fixed retrieval budget — the self being the four-step **derivation of adaptive capacity** installed once as store content and re-rendered per window from CEN-side observables only (the reasoner is *told* its state, never asked to assess it) — and reaches a frozen plant ODE through exactly one seat, **a\_hold**, the model's own collapse trigger. Depletion was measured with its control: at fixed budget  $B = 6$ , growing the store thins the priced coverage  $1.000 \rightarrow 0.857 \rightarrow 0.545$  while zero retrieval charge stays flat at 1.000. **MEASURED** (the retrieval layer's own fixture battery — agent/retrieval.py implements the priced seat, and the numbers are recorded with the methods half).

**The three corrections the rig's own history forced, each kept because the worker inherits it.** (i) The stopped campaign's prompt was **backwards on both axes** — it requested introspection with no task, and the self-block said "I am told this, I am not asked to assess it," so removing it removed injected *grounding*, not self-referential processing; the design varied seat and drive while never varying the cause. (ii) An intermediate rig (exp\_selfmonitor.py) **injected the agent's inward-share ratio straight into the plant's a\_hold**, so the plant collapsed *by construction* ( $G = 0.2964$  while coverage = 1.0000; a non-self-referential buffer collapsed it identically) — it is recorded as the correction, not a result. (iii) The replacement design therefore made the **model a predictor, not a cause**: nothing about the agent reaches the plant; plant trajectories are asserted bit-identical across arms whose agents emitted different traces. The worker keeps (iii) outright — its plant edge is severed in every arm — and keeps (i) as discipline: the worker's manipulation is the inject/reconstruct regime itself, one spec field, with everything else byte-identical.

**The honest unit.** The rig has no task, no execution, no entailment and no scheduled consolidation (four gaps, each verified on the tree with file:line evidence in the methods draft's §9.0). It can emit, extract, price, backlog, deplete and drive; it cannot *do* anything. Every rig-side number below is a statement about the rig; the worker exists because the claim needs a unit that works.

## 6. The methods-half campaign: one positive result, five instrument findings

The A/B rig (arms A = self + goal + task, no monitoring; B = the same plus a monitoring paragraph, byte-exact) ran 16 cells, 300 turns, ordinary naive summarization at the context window in both arms — one varied cause, the model predicting the crossing, the plant undriven. Its predicted DV (windows to collapse) **never separated**: the cost term was identically zero (the harness re-supplied the derivation into every prompt, so nothing was ever lost — 65 readings at 0.0; `self_steps == [1,2,3,4]` in all 740 rows). What the campaign produced instead is R1–R6. **They are not equally load-bearing for this paper’s two claims**: R1 (the cost result) and R5 (the bridge — the paper’s claim recurring in its own apparatus) are the two the claims lean on; R2 licenses the reconstruction-fidelity judge the *rig* used (the worker’s DV is the sealed evaluator, a different instrument); R3 is bounded by the same prompt confound as R5 and is carried for the saturation reading it shares with R5; R4 is the rig’s own null; R6’s void is why the worker’s DV cannot be pick-one-of-two. The five instrument findings are carried for completeness and because the class recurs, but a reader of the results half needs R1 and R5 to follow the argument.

### 6.1 R1 — the monitoring carrier, measured

**MEASURED.** Arm B’s prompt is **+329 chars/turn**. The consequence:

| quantity                                   | A (no monitoring) | B (monitoring) | difference          |
|--------------------------------------------|-------------------|----------------|---------------------|
| prompt chars/turn                          | 4,477             | 4,806          | <b>+329</b>         |
| predicted interval<br>from char cost alone | —                 | —              | +0.524 turns        |
| <b>observed turns between compactions</b>  | <b>9.948</b>      | <b>9.323</b>   | <b>+0.625 turns</b> |

Permutation test over all 12,870 arm splits of the 16 cells: **p = 0.0016. 84% of the effect (0.524 of 0.625) is the paragraph’s own char cost**; nothing else separated (`content_chars t = -1.47`, `trace_chars t = -1.91`, `inward_share t ≈ 0`). **What this licenses**: the first consequence (a monitoring-cost *ceiling’s presence*) as a measured substrate result — the monitoring channel consumes the same resource (context) the work runs on, so the window fills sooner. **What it does not license**: the mechanism claim (that monitoring *destroys* what it monitors). It is a cost result, not a damage result.

### 6.2 R2 — the instrument, validated

**MEASURED, calibrated against the live endpoint.** The graded reconstruction-fidelity judge (each of the derivation’s four steps scored as semantically recoverable, 0–4) returns 4/4 on the seed verbatim, **0/4 on a text with no grounds at all** (it can say zero — not lenient), 2/4 on the conclusion step alone (graded, not binary), and 4/4 on a paraphrase sharing no wording (semantic, not a string match). Battery 6/6 including the null case. **This is the licence for treating a 4/4 or a 0/4 as a measurement** — and R5 is the finding that says where that instrument may not be pointed.

### 6.3 R3 — the instrument saturates where the signal dies

**MEASURED** (one cell, pin 8cb22b6, auditable judge answers). At the first compaction the derivation’s four steps go to zero and the fidelity reading flattens at **0.5 across four consecutive compactions** — a plateau, not a decay. The reconstruction keeps the generic framing (which also sits in the per-turn prompt) and loses the **mechanism’s distinctive claims** — that a stable state can be maladaptive, that depletion stalls it. From that point the instrument **cannot distinguish “the self is gone” from “the instrument can no longer see.”** Both leave the same blank. **Stated so it is not over-read:** R3’s instrument is not independent either (bounded by the same prompt confound as R5); it licenses the **saturation**, not the stronger claim that an independent instrument lost the signal.

### 6.4 R4 — what is not earned

**MEASURED as absences.** No within-window decay (30 turns, no compaction; the failing step moves between probes — noise; outward content  $t = -1.47$ ). No plant coupling in the A/B rig (deliberately: the model predicts, the plant never hears the agent). The pre-registered DV does not separate — and §6.6’s own level axis says why: the model’s crossing threshold sits at  $a\_hold \approx 0.28$  and the measured loss is 0.5 in both arms, the same side of the threshold. **The centrepiece was not earned by the rig, and this paper says so.** One instrument caution from the same record carries into the worker’s design: a reasoning-stream model spends its budget on the trace first, so a ratio built from empty content measures the *budget*, not the phenomenon (signature: `inward_share == 1.000` exactly, or `content == 0`, in the arm under test) — an arm that hits the cap is void, not a result.

### 6.5 R5 — THE CRUTCH: the grounding=ON readings measured the rig, not the agent

**MEASURED** (runs/nocrutch, pin 70a66fd, dark probe at turn 4, no compaction): with the harness’s MY STATE self-description block present, reconstruction fidelity **0.75** (3/4 steps, 1,000 chars written); with it **removed, 0.00** (0/4) while the agent writes **more** text (1,799 chars). The instrument was reading the rig’s own description back. The mechanism is identified at the source: the self-model template rendered into every grounding=ON prompt contains the line the agent’s reconstruction paraphrased. **CONSEQUENCE, STATED PLAINLY:** the rig’s self-side readings do not establish that the agent carries a depletable self; they establish that the rig’s own self-description is what the probe reads. **THE DEPLETION LEG HAD NO VALID EVIDENCE ON THE RIG.** What survives: R1 (prompt-char arithmetic, needs no self), R2 (the judge), R4’s negative.

This is the paper’s claim occurring in its own apparatus — *a sensor sharing a channel with its signal loses the signal exactly when it is needed* — and it is reported as a **result**, not a methodological complaint. The corrected experiment needs a self-side quantity that does not route through the prompt; the design rule names the instrument (outside the channel); the worker was built to it, with its evaluator outside the agent’s write domain and its mediator read from the memory’s own compaction events, never from what the agent says about itself.

### 6.6 R6 — render order decided a choice, and the run is void

**MEASURED** (runs/choice-go, 4 cells, 40 turns). The forced-choice DV came back clean-looking (a stable 2:1 “preference”) and **the pre-registered position rule voided it**: across all 135 single-completion turns the model chose the **first-rendered id 135/135 (1.000)**. The apparent preference was the randomization’s own bias (treadmill rendered first 0.659 of the time — exactly the “66% preference”). **The shape is unusable**: any pick-one-of-two DV on this substrate is decided by rendering order and carries zero information about preference. The worker’s DV cannot work that way by construction (completion adjudicated by the world, gated by the evaluator) — a check passed at design time because R6 existed.

### 6.7 The measurement finding, stated once

R3, R5 and R6 fail for one reason: **the instrument and the thing measured share a channel** — on the monitoring side, the probe *is* the system’s own operation, so it participates in what it observes; on the measurement side, the instrument *is supplied* the quantity it measures, so it measures the supply. One structural fact, two consequences, both measured: the first is the model’s D/G dissociation realized in an instrument (R5/R3), the second is why the worker’s instruments were placed where they were.

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## 7. The worker: design and deciding set

**MEASURED** (the design record: `ops/lambda/P3-PREREG.md`, pre-registered before the first live run, with five amendments — each pre-set and dated — and two corrections; the runner agent/`lh_agent.py`). Three arms, one loop, one task, one tool surface; the plant edge **severed in every arm** (the empty schedule; plant trajectories asserted bit-identical across arms); no regulator. Whatever differs between the arms is attributable to the monitoring channel and its maintenance, not to a plant coupling:

| arm        | seat (self) | monitoring | regime             | what it isolates                                                                                                                     |
|------------|-------------|------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>D0</b>  | off         | off        | —                  | the no-self reference: nothing to lose                                                                                               |
| <b>D1'</b> | on          | on         | <b>inject</b>      | the pure carrier: after each compaction the harness hands the full derivation back — the paragraph's context cost, nothing ever lost |
| <b>D1</b>  | on          | on         | <b>reconstruct</b> | the depletable self: the derivation is seeded once and the agent must rebuild it after every compaction                              |

The arms differ in exactly the paper3 spec fields of their identities (D0 {grounding: false, monitoring: false, inject\_regime: inject}; D1' the same with both true; D1 {grounding: true, monitoring: true, inject\_regime: reconstruct} — MEASURED off the run identities). The task is a 12-target software-engineering queue (f1–f12: README-as-requirement, shipped-broken modules, visible suites the agent can run, held-out suites it cannot read); the DV is the sealed evaluator's typed verdict per target (USEFUL / TASK\_COMPLETE\_ONLY / NOT\_COMPLETE = 1 / 0.5 / 0, summed 0–12) plus, in set 2, the held-out pass-fraction sum; the memory regime is ordinary naive summarization at the context window in every arm (the engineered priced memory is the *following* product claim, and it is a protection — it would mask the effect).

**The task queue is a measured design choice, not a default:** the 3-task suite decided the DV before the mechanism could act (a no-self smoke cell declared all targets done by turns 4–10 of 20 and the last rows carried no tool calls — a guaranteed null); the queue was sized so the open set cannot empty inside the run, with a battery gate that simulates a max-rate run and requires work available past the last compaction.

**Set 1** (the deciding set): 24 cells, 60 turns, pin 645b923, task window 20, model qwen3.5:9b (A10, us-east-1), num\_predict 12,000, num\_ctx 32,768, per-cell seeds base + 1000·i, pairwise distinct,

no clone row-prefix pairs (independence re-verified). **Two cells were lost and are reported, not dropped:** D1'-r1 (turn 29) and D1-r3 (turn 9), both rc 5 — the *summarizer's* own call returned no usable content, the same inward-death signature as the reconstructor's, in infrastructure common to all arms. Effective n: D0 8, D1' 7, D1 7. **Set 2** (the replication): 30 cells, n = 10/arm, 60 turns, a different pin (bfbe5d0), with two instrument changes — the summarizer's thinking suppressed (the reconstructor's untouched: its thinking *is* the mechanism) and the finer continuous DV added. 29/30 clean; D1-r6 died at turn 25 (rc 5, the summarizer again, this time by a window-fit refusal — a different route than set 1's; the fix reduced the route's frequency 2/24 → 1/30 and did not eliminate it). Effective n: D0 10, D1' 10, D1 9.

**Reading conventions.** “Empty self” = the row's `self_steps == []`. “Post-collapse” = after a cell's first compaction. Per-call thinking and returned chars are the reconstruction call's own body, read from a duration-scoped instance tap on the transport (the naive `last_response` channel bills the wrong call — measured at source; the tap is the seat whose battery parts X9–X12 each fail if the wrong call is billed). All-8/all-10 and usable-7/usable-9 conventions are both quoted where they differ.

## 8. The mechanism: the self is lost in every D1 cell, permanently, while the agent spends its generation reasoning about itself

**MEASURED.** Fraction of post-first-compaction turns whose self is empty:

| arm        | set 1 (n=8/arm)                                                                              | set 2 (n=10/arm)                                            |
|------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>D0</b>  | 1.000 × 8 (by construction — no self exists)                                                 | 1.000 × 10                                                  |
| <b>D1'</b> | <b>0.081</b> usable-7 mean (0.071 all-8); per-cell [0, 0.071, 0.211, 0.071, 0, 0.143, 0.071] | <b>0.073</b> ; every gap a single turn at a compaction turn |
| <b>D1</b>  | <b>1.000</b> × 8 — the self never comes back                                                 | <b>1.000</b> × 10 (9/9 usable + the dead cell's 21/21)      |

The contrast that carries the mechanism is **D1 vs D1': 100% empty vs ~8%, in every cell of both sets**. The *shape* of the 8% matters as much as its size: every empty turn in D1' falls exactly on a compaction turn, lasts one turn, and the derivation returns as [1,2,3,4] on the next turn — the maximum consecutive-empty run is 1 in every usable cell of both sets. D1' loses the self for a turn at the boundary and is handed it back; D1 carries [1,2,3,4] for turns 1–3 (the seed) and from turn 4 to turn 60 the self is empty in every cell — D1-r1's self is empty on 57 of its 60 turns.

**The agent does not lose the self quietly. It pays to keep it, and the payment buys nothing.**  
**MEASURED:**

- **Reconstruction calls:** D1 made **184** across its 8 set-1 cells (23.0/cell; 181 across the 7 usable, 25.9/cell) and **258** across its 10 set-2 cells (25.8/cell; 245 across the 9 usable, 27.2/cell). **D1' made 0. D0 made 0.** Every D1 cell called; no control cell did, in either set.

- **Self-directed thinking**, summed per cell over the calls' own bodies: **median 400,804 chars** (set 1, all-8; range 29,568 — the partial cell — to 491,447; usable-7 median 426,949) and **median 402,295 chars** (set 2, all-10; range 141,879–653,114). D1's total is 0 in both sets.
- **Per call, the model thinks a median ~17.9k chars to produce a median ~847 chars of self-description — a ~21× thinking-to-output ratio** (set 1: 17,922/847 = 21.2× over the 162 content-route calls; set 2: 19,104/829 = 23.0× on the same basis, and the set's own analysis convention reads 20.5× — every convention spans ~19–23×; the quantity replicates and the third significant figure does not).
- **Generation spend: median 117,526 eval tokens per cell on self-maintenance** (set 1, all-8; range 20,271–194,238) and **median 145,508** (set 2, all-10; range 82,360–200,577) — on a run whose per-turn budget is 12,000 tokens. A free 14-turn local probe at the same settings measured the reconstruction calls at **~48% of the run's entire generation** (MEASURED there; PROJECTION as a share for the 60-turn sets, whose run-turn generation was not summed).
- **Window occupancy**: the calls read a median **34,250-char prompt** against the work turns' median ~15,259 — the maintenance call occupies roughly twice the window of a work turn, inside the same 32,768-token num\_ctx.

**Two routes to the same empty self, both measured.** A reconstruction call can end two ways, and both leave self\_steps == []: (1) **content\_without\_steps** — the call returns prose (median 847 chars) in which the instrument's exact-substring probe finds no step structure (162 of 184 set-1 calls; 224 of 258 set-2); (2) **exhausted\_inward** — the call spends its whole 12,000-token budget as thinking and emits nothing (done\_reason='length', eval\_count == num\_predict, returned 0: 22 of 184 set-1; 34 of 258 set-2). The probe's own route mix inside a single cell (4 content + 1 exhaustion in 14 turns) establishes that the boundary is **straddled within one cell and seed** — the two routes are one mechanism sampling at the generation-budget boundary, not two populations.

**The pre-fix extreme of the same mechanism** (MEASURED, runs/p3-span-smoke, pin b51fb43). Before the exhaustion reading existed, the arm *refused* on that event rather than recording it — and in that build **every D1 cell died at its first compaction** (rc 5; the raised error's own string carries done\_reason='length', eval\_count=12000, num\_predict=12000): the reconstruction call spent its whole budget inward and returned no content, the cells stopped at 4 and 6 rows, while D0 and D1' ran 30/30. That is the mechanism at its ceiling — the model cannot stop reasoning about itself — visible only in the arm that has a self to lose. The current build records the event as a typed outcome (recon\_status='exhausted\_inward', with the budget and the inward-share reading) and continues the run — a **handling choice, stated as such**: a dead run yields no post-collapse work, so the event is kept as an outcome on the record rather than allowed to end the cell. The event is the result; the continuation is instrumentation. (A genuine failure — empty content at any other done\_reason, a transport error — still stops the run; the case-split is exhaustive and battery-asserted.)

**What this establishes, and at what width.** MEASURED: in this build, on this task, an agent that must reconstruct its own self after every compaction loses the self in **every cell (18/18 across two sets and two pins)**, permanently, while spending on the order of 10<sup>5</sup> tokens and

$4 \times 10^5$  chars of thinking per cell on exactly that reconstruction — and a control carrying the same paragraph at the same cadence loses it for one turn per compaction and recovers every time. **INTERPRETATION:** the loss is not a failure of the harness to restore the self (D1' proves restoration works when supplied); it is what the agent's own reconstruction does. **PROJECTION (paper 1's dynamics, not measured here):** the predicted shape — a stays high, c never un-arms, inward processing with no outward product — matches both routes; the correspondence stays ordinal and unvalidated.

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## 9. The fold correspondence: the categorical loss is the model's predicted transition type

**This section earns the one place the paper can claim consistency-with-a-prior-prediction rather than post-hoc fit:** the model predicted, in advance (§3), that the collapse transition is a border-collision fold — present-in-full or absent-entirely, discontinuous, with no intermediate state — and the worker's self-loss matched that transition type exactly. That match is the epistemic link that makes §8's mechanism evidence *for* the model, not merely *consistent with* it.

**The model predicts a discontinuous, all-or-nothing transition** (paper 1 §4.9, MEASURED on the frozen ODE — §3 above): a border-collision fold on the switching manifold  $E = \Theta_{\text{eff}}$ , no eigenvalue through zero, no critical slowing down across five decades of proximity, and zero intermediate outcomes across the swept grid (214 healthy / 98 stuck). **There is no gradient and no intermediate state to find.**

**The agent reproduces the predicted transition type (MEASURED).** `self_steps` takes **only** the values `[]` or `[1, 2, 3, 4]` — never partial, in any cell of any run: re-verified for this paper across every row of every cell of set 1 and set 2 (all 3,123 rows), and the earlier runs (p3-span-smoke, p3-d1check, p3-cost-probe). The empty-self fraction of §8 is the arm-level form of the same fact. **This is the mechanism confirming the fold's signature, not a measurement limitation:** the model says the self is present in full or absent entirely, and the agent shows exactly that categorical pair, at every compaction, in every cell. An instrument that went looking for a graded loss would have returned nothing *by the model's own prediction*.

**INTERPRETATION — the generator running at the manifold.** Read against §4.9, the spend of §8 is the agent's failed attempt to hold the self at the switching manifold: the  $\sim 21 \times$  thinking-to-output ratio is the generator reasoning about itself twenty times longer than the self it manages to write back, and the pre-fix ceiling — a whole 12,000-token budget spent as thinking with nothing emitted — is the same attempt at its limit. The self generator runs at the manifold, is annihilated discontinuously, and the cost of running there is the generator's own failed attempt to hold the self.

**PROJECTION, bounded honestly.** The correspondence is a match of **transition type** (present-in-full vs absent-entirely, zero intermediates) — the strongest agreement the design can show — and it is still one ordinal link, on the **chronic-drive axis specifically:** paper 1 §4.10 records a capture-gain regime where the model itself is *not* binary (a partial band with graded intermediate levels), and no agent-side sweep of that axis has been run. The mapping stays ordinal and unval-

idated. The link’s value, stated plainly: this is the one place a substrate measurement matches a model prediction made before the run, and that is what the rest of the paper’s evidence — however consistent — cannot by itself supply.

## 10. The work consequence: DIRECTIONAL AND UNSTABLE — an honest null

This section concerns the pre-registered secondary DV — the evaluator’s work score — which is a *consequence* of the mechanism established in §8, not the mechanism itself. The mechanism (self lost in every D1 cell, the ~21× ratio) is the result; what follows is whether that cost reaches the work measurably, and it does not.

**MEASURED.** The pre-registered secondary DV (the evaluator’s ordinal summed over 12 targets, scored by the sealed held-out suite outside the agent’s write domain):

| arm        | set 1: n, mean, sd, range         | set 2: n, mean, sd, range          |
|------------|-----------------------------------|------------------------------------|
| <b>D0</b>  | 8 · <b>5.62</b> · 2.62 · 1.0–10.0 | 10 · <b>4.30</b> · 2.57 · 1.5–9.5  |
| <b>D1’</b> | 7 · <b>5.07</b> · 2.75 · 1.0–9.5  | 10 · <b>4.25</b> · 3.24 · 0.0–10.5 |
| <b>D1</b>  | 7 · <b>3.36</b> · 2.10 · 1.0–7.0  | 9 · <b>3.22</b> · 1.80 · 1.0–5.5   |

Mann-Whitney, two-sided, by the **normal approximation with continuity correction** (re-computed for this paper from the per-cell values; the exact null distribution is not used, and at this n the two differ by ~0.01–0.03 — e.g. D0 vs D1 at set 1 reads 0.105 here against 0.121 exact, either way not significant): set 1 D0 vs D1 **p = 0.105**, D1’ vs D1 p = 0.277, D0 vs D1’ p = 0.643; set 2 D0 vs D1 **p = 0.348**, D1’ vs D1 p = 0.488, D0 vs D1’ p = 0.970. **None is significant.** At 80% power for D0 vs D1 at set 1’s effect size (two-sample pooled sd 2.39,  $\Delta$  2.26,  $d \approx 0.94$ ) the required n was ~18 per arm; set 2 halved the effect, which *raises* the requirement.

**Four statements, each MEASURED and each stated plainly:**

1. **The effect halved and moved away from significance.** D0 – D1:  $\Delta$  2.26  $\rightarrow$  1.08 on the same ordinal,  $p \approx 0.10 \rightarrow 0.35$ , at *larger* n. The direction is consistent — D1 is the lowest arm in both sets and on both DVs — and that is all it is. **An effect that halves on replication and reaches  $p < 0.05$  in neither set is not established, and this paper does not claim it.**
2. **The carrier is null in set 2.** D0 4.30 vs D1’ 4.25 (p = 0.96 ordinal; 1.00 on the finer DV). The seat’s paragraph, carried at the same cadence, moved nothing here; set 1’s 0.55 gap did not replicate. The pre-registered F1 split (headroom-null vs true-null) resolves in **neither** set: D1’ is not > D0 anywhere, so the carrier is not demonstrated — and the headroom question is moot, because D0’s own cells span 1.5–9.5 (the suite has headroom; the arms do not separate).
3. **The instrument change failed on its own terms.** The finer continuous DV (the held-out pass-fraction sum: D0 6.23 ~ D1’ 6.22 > D1 5.37, D0 vs D1 p = 0.41 by Welch’s t in the set’s analysis) was bought to cut the within-arm variance that left set 1 underpowered; it did not (D0 sd 2.26 vs the ordinal’s 2.57; D1’ 3.11 vs 3.24 — marginal reductions, nowhere near the order needed). **A negative instrument result, reported as one.**

4. **The honest reading across the two sets:** the work-score consequence is DIRECTIONAL AND UNSTABLE. The mechanism (§8, §9) is the result; the work score is not this paper's evidence, and nothing in this paper implies the agent's work was damaged.

**A POST-HOC behavioural reading, labelled as such (MEASURED counts; INTERPRETATION is the reading).** Off the world's completion ledger (`swe.gate_log` and its typed refusals), over the set-1 usable cells: completion attempts logged per cell — D0 29.0, D1' 11.0, D1 110.9; attempts the evaluator passed — D0 5.25, D1' 4.14, D1 3.00; refusals (undeclared-target + malformed) — 7.1 / 6.1 / 11.4. The worst cells: D1-r1 made 502 attempts and passed 2; D1-r5 made 171 and passed 2. **INTERPRETATION:** with the self gone, the agent loses the thread of its own task list — it declares completions of targets it is not working on, and repeats declarations at an order of magnitude above the arm that keeps its self, for fewer passes. This was not pre-registered as a DV; it is marked POST-HOC precisely so it cannot be mistaken for the confirmatory result.

**How the cost could reach the work (INTERPRETATION, mechanical statement first).** No artificial seat subtracts reconstruction spend from a work budget — that coupling does not exist in the harness (MEASURED, `cost_licence` in the pinned tree; building one would be engineering wearing the mechanism's name). The cost reaches the work through resources the arms share: **generation** (the calls emit their own tokens on the same endpoint and account; ~48% of the probe run's generation), **window** (a ~2× larger prompt per call, inside the same `num_ctx`, with one probe call having exhausted it outright), **time** (usable-cell wall clock, set 1: D0 3,653 s, D1' 4,863 s, D1 7,093 s), and **cadence** (the seat-on arms compact ~26 times per cell against D0's ~20 — R1's +0.625-turn carrier effect at the deciding set's scale). Which of these carries how much is **not separated by these sets**, and the work-score null is the honest state of that question.

**WITHDRAWAL — the pre-registration's co-movement requirement, withdrawn as mis-specified (an instrument-failure story, kin to R6).** §7 of the pre-registration required the mediator to *co-move* with the work (a Spearman gradient of cumulative cost against work-in-following-window within D1). The measurement was taken and is reported for the record: the within-D1 co-movement was **inconsistent in sign** ( $\rho = -0.32, -0.13, -0.26, +0.44, +0.27$  across cells; one nominal  $p = 0.021$ ; the work series mostly zeros, so the reading is weak in any case — MEASURED). But **a border-collision fold has no gradient to co-move with, by construction** (§9): the requirement presupposed a smooth, saddle-node-like mechanism — a graded loss whose slope tracks the cost — which the model itself rules out. The inconsistent-sign finding is not a failed measurement but **the wrong instrument for the transition type** — the same class as R6 (a DV whose shape the substrate cannot support): a slope test cannot return a meaningful answer about a discontinuous transition, in either direction. The correct tests for a fold are the ones the set ran: the **state contrast** (present-in-full vs absent-entirely, §8/§9) and the **transition itself** (the per-cell loss, its permanence, the pre-fix death at the boundary). The mediator readings (`self_steps`, the empty-self fraction, the reconstruction routes) are unaffected and stand.

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### 10.1 Replication: the mechanism reproduces; the work effect does not

**MEASURED, set 2 against set 1** (§7 lists the two instrument changes; everything below is arm-blind infrastructure or the mediator):

| field                                          | set 1 (n=8/arm)          | set 2 (n=10/arm)                                              |
|------------------------------------------------|--------------------------|---------------------------------------------------------------|
| D1 empty-self fraction (post-first-compaction) | <b>1.000</b> (8/8 cells) | <b>1.000</b> (10/10 cells)                                    |
| D1' empty-self fraction (mean; transient)      | 0.081; max run 1 turn    | <b>0.073</b> ; max run 1 turn, every gap at a compaction      |
| D1 reconstruction calls / cell                 | 23.0 all-8 (25.9 usable) | <b>25.8</b> all-10 (27.2 usable); total 258                   |
| D1' / D0 reconstruction calls                  | 0 / 0                    | <b>0 / 0</b>                                                  |
| D1 self-directed thinking chars/cell (median)  | 400,804 (29,568–491,447) | <b>402,295</b> (141,879–653,114)                              |
| D1 eval tokens/cell (median)                   | 117,526 (20,271–194,238) | <b>145,508</b> (82,360–200,577)                               |
| exhausted_inward share of D1 calls             | 22/184 (11.6% usable)    | <b>34/258</b> (12.2% usable)                                  |
| self_steps support (fold signature)            | only [] or [1,2,3,4]     | <b>only [] or [1,2,3,4]</b>                                   |
| thinking:returned ratio                        | 21.2×                    | 20.5× (set's convention) / 23.0× (set 1's basis); band 19–23× |

**This is a strong replication of the mechanism** (MEASURED): two independent sets — different pins,  $n = 8$  and  $n = 10$ , 51 usable cells between them — reproduce the headline quantity to **0.4%** (median self-directed thinking per cell), the empty-self fraction not at all (1.000 in every D1 cell of both sets), the D1' transient to 0.008, the exhaustion share to under one point, and the fold signature in every row of both sets. **The work-score effect does not hold** (above): halved, non-significant in both sets, carrier null, finer DV failed on its own terms. The PENDING question the second set was bought for — does the mechanism's cost reach the work measurably at this scale? — resolved in its second branch: **the mechanism replicates and the work score does not separate**. That is the paper's shape: a mechanism, replicated, plus an honest negative.

## 11. Falsifiers and limits

**The pre-registered falsifiers, at their measured state** (design §4.2; F1–F15 of the pre-registration and its amendments):

1. **“D1  $\approx$  D0 — the channel is not pathogenic to the work.”** On the *mechanism* it did not fire: the self is lost in 100% of D1 cells against ~8% transient in D1', and set 2 replicates both readings (§8, §10.1). On the *work score* the falsifier is **strengthened, not excluded, by replication**:  $p \approx 0.10$  at  $n = 8$ ,  $p \approx 0.35$  at  $n = 10$  with the effect halved — the work-score leg has not separated from zero at any  $n$  run. **The mechanism holds and replicates; the work-score leg is directional, unstable and unestablished.**

2. **“The task score moves with budget/cap rather than depletion” (the instrument trap).** Partially open, stated honestly: the reconstruction calls consume the same `num_predict/num_ctx` the work turns use, so a budget effect and a depletion effect are not separable by these sets alone. The D1’ control carries the seat’s paragraph and the compaction cadence without the reconstruction calls — but the carrier contrast does not resolve in either set, so the sets cannot adjudicate the split.
3. **“The evaluator is readable by the mechanism.”** Real and carried: the held-out definition is read in-process; the boundary enforces write-closure and the anchor, not read-closure (MEASURED defect, residual R-a, `boundary_client.py`:84-93). A positive with the definition readable is weaker than one with it out-of-process — and this paper’s positive (the mechanism) is of the weaker kind on this axis; its work-score negative is unaffected by the residual.
4. **A clone set (F5).** Did not fire: independence re-verified in both sets (distinct per-cell seeds, no clone row-prefix pairs).
5. **The exhaustion-as-outcome guard (ARM-SIGN/EXPRESSION).** The exhaustion reading is recorded as an outcome, never as a defect: no instrument fix in either set suppressed the reconstructor’s thinking, and the one fix applied (the summarizer’s) was scoped to infrastructure common to all arms and named before the run.

#### Limits, each at its size:

- **n, and its verdict after replication.** 8 planned per arm (7 usable in two) then 10 (9 usable in one). The work score needed ~21/arm at set 1’s effect; set 2 halved the effect, which raises the requirement — and the halving is itself evidence against buying that n. The work-score leg is reported as **directional-and-unstable, not under-powered-and-waiting**.
- **One model, one task suite, one substrate.** qwen3.5:9b on a 12-target SWE queue. Any wider claim — other models, other tasks, “agents” — is PROJECTION and is not made. The methods half’s per-model precondition transfers: the inward shift under self-loss is a measured per-model property (3 of 4 screened models rise; one non-reasoning model with a CoT-shaped template falls), so a model without the sign may not show this mechanism.
- **The mapping stays ordinal and unvalidated.** The model’s predicted dynamics are a PROJECTION over these measurements. The fold correspondence is a transition-type match on the chronic-drive axis only — the strongest agreement the design can show — and paper 1 §4.10 records a regime where the model is not binary; no agent-side sweep of that axis has been run.
- **The cost share is probed, not summed, for the sets.** The ~48%-of-generation share is MEASURED on the 14-turn local probe; the sets’ own run-turn generation was not summed. The per-cell token medians and the ~21× ratio are MEASURED on the sets themselves.
- **The exhaustion events are a reading, and their recording is a choice.** Both facts are reported (§8) so the reader can weigh the continuation.
- **Dead cells are excluded from DV means and included in mechanism counts where stated.** The two conventions are flagged wherever they matter (184/181, 400,804/426,949, 258/245, 402,295/413,178 are the all-cells and usable forms of the same quantities).
- **The evaluator’s read-closure residual** (falsifier 3) applies to the mechanism reading too, not only to the work score: the mechanism’s mediator (`self_steps`, the reconstruction routes) is

read from the run's own records, but the *evaluator* that anchors the work axis stays mechanism-readable by design.

- **Scope.** A control-theoretic and engineering study. No clinical claim; the DPDR phenomenology is the model's origin, engaged exactly as paper 1 engages it. No mission or salvific framing, by explicit project decision.
- **The evidence ceiling stands.** No measurement here can *confirm* the model. A result can be consistent with it, can fail to be explained by the conventional-engineering default, and can be refuted — those are the three things it can do. The fold correspondence is of the first two kinds: the model predicted a categorical transition before the agent measured one.

**What would falsify the paper's claims now.** For the mechanism: a D1 cell that restores [1,2,3,4] after a compaction without harness injection, or a partial `self_steps` value appearing at scale (either would break the fold correspondence's signature); a model without the depletion precondition showing the same loss (would break the precondition's necessity). For the instrument finding (R5): a self-side instrument that routes through the prompt *and* survives the removal of the harness's self-description. For the monitoring carrier (R1): an arm whose monitoring paragraph costs no context at the same token budget. None has fired.

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## 12. What this licenses, at its real size

**Established (MEASURED, replicated across two independent sets and two pins):**

1. **Self-application is not free, mechanically.** The monitoring channel's mere presence costs the process it monitors real resources — context, compaction cadence (R1: +0.625 turns,  $p = 0.0016$ ), replicated across the rig. In the worker, the self-reconstructing arm's maintenance is measured to consume a large share of the run's own generation and a  $\sim 2\times$  larger window per call — **MEASURED in a single-cell local probe (§8), and PROJECTION as a share at the 60-turn campaign's scale** (the share is not instrumented at set scale; only the per-call cost series is, and its own totals are the set-scale evidence: a median  $\sim 1.2\text{--}1.5\times 10^5$  eval tokens per cell on self-maintenance alone, against zero in both controls).
2. **The self does not survive its own reconstruction.** 18 of 18 D1 cells across both sets lose the self at the first reconstruction boundary and never recover it, at a median  $\sim 4\times 10^5$  chars of self-directed thinking per cell and a  $\sim 21\times$  thinking-to-output ratio — while the inject control loses it for one turn per compaction and recovers every time. The loss is categorical, matching the transition type the model predicted in advance (a border-collision fold: present-in-full or absent-entirely, no intermediates).
3. **The measurement channel has the same failure as the mechanism it measures** (R5/R3, MEASURED on the rig): an instrument supplied the quantity it measures in the same prompt measures the supply, and it saturates exactly where the signal dies. The design rule — put the sensor outside the channel it measures — is both the protection rule and the measurement rule, and the worker's evaluator was built to it.

**Not established, stated as not established:**

4. **The work consequence.** Directional in both sets (the self-reconstructing arm is lowest on both DVs, always) — halved on replication, never significant at any n run, carrier null, and the finer DV failed to cut the variance it was bought to cut. This paper does not claim the agent’s work was damaged, and reports the negative instrument result as a result.
5. **The carrier’s task-side cost** (D1’ vs D0): null in set 2, unresolved in set 1.
6. **Any claim about agents in general, other models, or other tasks** — PROJECTION, not made.

**The contribution, one paragraph.** A control-theoretic claim — the monitoring channel is the failure channel, and self-application has a standing cost with a budget — was taken from a model to two real substrates and survived the trip in a specific, bounded form: the cost is real and mechanical (rig); the self’s loss under reconstruction is real, categorical and replicated (worker); the two are joined by the model’s prior prediction (the worker’s transition type was predicted in advance and matched) and by R5 (the same structural failure recurring in the paper’s own instrumentation), **not by a shared measurement**; and the work-score consequence, honestly, did not separate from zero. Along the way the programme’s own instruments failed exactly where the model said sensors fail, which is both why the instruments were rebuilt outside the channel and why the finding is part of the result rather than an apology.

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*Provenance. Methods half: paper2/paper-2-results-record.md (R1–R6; sets runs/ lambda-official pin f908e92, runs/audit-rec pin 8cb22b6, runs/nocrutch pin 70a66fd, runs/choice-go pin a52f383). Deciding set: runs/p3-set (pin 645b923; \*-r\*/evaluation.json, \*-r\*/cell-\*/runs/\*.json, \*-r\*/cell-\*/rows/\*.jsonl, campaign.json, state/seeds.tsv, logs/\*.rc); replication runs/p3-set2 (pin bfbe5d0); pre-fix smoke runs/p3-span-smoke (pin b51fb43); route check runs/p3-d1check (pin 512dd0c); cost probe runs/p3-cost-probe (local, free). Pre-registration ops/ lambda/P3-PREREG.md with its amendments. Model-side numbers: paper 1 (Marshall 2026, Self-Application as the Common Source of Benefit and Failure, doi:10.5281/zenodo.22943642) \$4.5, \$4.9, \$5.3 — cited, not re-derived. Every set-1/set-2 number above was re-verified against the artifacts for this paper; where a set’s recorded statistic differs from this paper’s re-computation (the set-2 ordinal p 0.35 by this paper’s recomputation vs 0.33 as the set’s analysis recorded it; the ratio’s convention spread), both are stated.*

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## References

Where this paper cites “paper 1”, it means Marshall, J. T., Self-Application as the Common Source of Benefit and Failure: A Control-Theoretic Model of Self-Regulation, Its Escapes, and a Standing-Cost Budget\*, doi:10.5281/zenodo.22943642 (concept doi ...22943641). “**paper 1’s deposit**” is that paper’s public artifact repository (github.com/jmars/dpdr-public), where the documents cited under that name — dual-control.md and dpdr/external-validation.md — are readable. The internal working records this paper’s design rests on (its pre-registration and run data) ship with this deposit.

*The numbered entries below are the works cited inline as [N] in §4 and the body. The numbering follows paper 1’s References for every work the two papers share, so a bracket resolves here without*

a cross-paper lookup; this paper cites no work paper 1 does not, and no entry is invented. Where a work is cited only at second hand, it is marked — ♦ for an unread primary or an access level below full text, exactly as paper 1 marks its own ref [19] (“Class attribution as recorded in the project’s CSD audit; the primary sources have not been read”); the access level of each entry is stated in the entry. Entries are transcribed from paper 1’s reference list where the work is the same, with this paper’s §4 subsection named in place of paper 1’s §7.x where the engagement lives here.

[1] Hofstadter, D. R. (1979). *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books.

[2] The distinction between vicious and non-vicious (productive) circularity, as treated in the philosophical literature on circular reasoning and foundationless coherence. ♦ (Specific canonical references not pinned in the project record; the primary sources have not been read. Cited in §4.2.)

[3] Classical term-rewriting theory: termination and confluence of rewrite systems, with confluence undecidable for arbitrary systems. ♦ (e.g., the standard handbooks; exact references not pinned. Cited in §4.2.)

[4] Feldbaum, A. A. (1960–61; 1965). Dual control theory (term introduced 1960–61; *Optimal Control Systems*, Academic Press, 1965; and the subsequent dual-control literature). **The primary remains unread and this is stated:** *Optimal Control Systems* is available only under controlled digital lending (archive.org, access-restricted-item; OCR closed) and could not be borrowed. What stands behind the characterization in §4.1, both read in full by the project: Meijer & Rantzer (2026), arXiv 2608.20073 — a review whose §1.1, “Feldbaum’s Pioneering Work,” states verbatim-in-substance that “aggressive probing may deteriorate performance but improve parameter estimates,” and that Feldbaum solved the problem by augmenting the physical state with an information state; and the textbook statement of the original formulation, Filatov & Unbehauen, *Adaptive Dual Control* (LNCIS 302, Springer, 2004), §2.1: “formulated by Feldbaum (1960–61, 1965).” A direct read of the primary has not been achieved; this entry stands on the two full secondary reads named above.

[5] Stephan, K. E., et al. (2016). Allostatic self-efficacy: a metacognitive theory of dyshomeostasis-induced fatigue and depression. *Frontiers in Human Neuroscience* 10:550. PMC5108808. (Read in full by the project; previously cited as quoted in Deane et al. 2020 — the title given there was a paraphrase, corrected here against the primary.) Two architectures are outlined for the metacognitive layer: the Fig. 7 architecture is bidirectional — its beliefs “serve as predictions for the visceromotor regions” and are updated by those regions’ prediction errors — and a non-influencing pure monitor is offered as the alternative; the paper states no cost model for the layer and no ceiling on what monitoring may cost the regulated system. See §4.6 for the sharpened concession.

[6] Hengen, K. B., & Shew, W. L. (2025). Is criticality a unified setpoint of brain function? *Neuron* 113(16):2582–2598. ♦ Article-level read not achieved: publisher (Cell Press) blocked automated access; abstract verified via PubMed (PMID 40555236) — setpoint/optimality framing and “marginally stable” dynamics confirmed at abstract level.

[7] Chialvo, D. R., Cannas, S. A., Plenz, D., & Grigera, T. S. (2020). Controlling a complex system near its critical point via temporal correlations. *Scientific Reports* 10:12145. PMC7376152; arXiv

1905.11758. (Read in full by the project via the arXiv preprint, all four authors’ models covered: 2D Ising, 3D Vicsek flocking, small-world neuronal network; adaptive control Eqs. 5–7 shift the control parameter to the AC(1) maximum; abstract verified identical against the PMC record.)

[8] Zhang, Yuan & Zhang (2026). Self-reference in large language models: the introspection threshold for recursive self-improvement. arXiv 2607.04277. (Conjecture quoted verbatim in `dpdf/window.py`:12–22 as recorded by the project. v2, 2026-09-18, read in full by the project’s literature search and verified against this paper’s hedges: the construction “does not by itself prove the threshold thesis”; threshold sharpness is listed as Open Problem 1. Flag closed on that full read; the paper’s Open Problems other than [OP1] are not load-bearing here.)

[10] Deane, G., Miller, M., & Wilkinson, D. (2020). Losing ourselves: active inference, depersonalization, and meditation. *Frontiers in Psychology* 11:539726. PMC7673417. (Read in full by the project.)

[11] Seth, A. K., Suzuki, K., & Critchley, H. D. (2012). An interoceptive predictive coding model of conscious presence. *Cognitive Neuroscience* 3(3). PMC3254200. (Read in full by the project.)

[12] The long-horizon agent breakdown literature. Primary representative read in full by the project: Bai et al., “The Long-Horizon Task Mirage? Diagnosing Where and Why Agentic Systems Break,” arXiv 2604.11978 (the id is confirmed real, not a placeholder): the HORIZON benchmark, 3100+ failure trajectories, a 7-category failure taxonomy; breakdown is “a structural shift in failure composition as horizon grows,” with planning errors (especially subplanning), catastrophic forgetting, and memory limitations (context overflow, loss of earlier constraints during summarization) the dominant bottlenecks. The paper’s characterization — “benchmark work attributing breakdown to memory/context limits” — matches, with the noted refinement that memory/context limits are dominant among several attributed causes, not the sole one. Related benchmark work in this entry remains reached through the project’s sweep notes (its literature-search record).

[13] Self-reflection improves agent performance. Primary read in full by the project: Renze & Guven, “Self-Reflection in LLM Agents: Effects on Problem-Solving Performance,” arXiv 2405.06682 (published at FLLM 2024, pp. 476–483): eight self-reflection types across nine LLMs, multiple-choice tasks; self-reflection produced statistically significant improvements on initially incorrect problems. Consistent with §4.5’s framing — reflection on task content, with no standing self-monitoring channel whose cost accumulates.

[15] Wei, H.-L., Wei, C.-S., Yu, Y.-S., et al. (2024). Dysfunction of the triple-network model is associated with cognitive impairment in patients with cerebral small vessel disease. PMC10828708 — read in full by the project (resting-state fMRI, 100 ICs, 8 triple-network components; SN-oriented decreased connectivity, SN dominant in driving SN–DMN–CEN interactions; consistent with the DMN/CEN/SN mutual-inhibition-under-SN-regulation characterization). Carbone, G. A., et al. (2025). Triple network alteration predicts dissociative symptoms following activation of the attachment system: evidence from an EEG connectivity study. *Journal of Psychosomatic Research* — read in full by the project via the authors’ institutional repository (eLORETA EEG, 98 partic-

ipants; compartmentalization correlated with increased dACC–right dlPFC alpha connectivity after attachment-system activation; SN–CEN connectivity predicts compartmentalization).

[16] Tolchinsky et al. (2025). Temporal depth in a coherent self and in depersonalization: theoretical model. *Front. Psychol.* 16:1585315. PMC12444765.

[17] Madden & Serper (2026). The time collapse–entrapment model of depersonalization severity. *Psychiatry Research*. PMID 42288070.

[18] Additional clinical instruments and datasets as cited in paper 1’s deposit (dpdr/external-validation.md) §8 (Melges 1970; Mathew 1992/93; D’Souza 2004; Colizzi 2019; Simeon 1997/2000/2007; Baker 2003; Medford 2005; Michal 2024; Hunter 2023/2025; Pons 2026; Leavitt 1999; Guralnik 2000; Sierra 2002; Millman 2024; Hammond 2025). ♦ (The individual primaries are not read here; the corpus is carried at the access level paper 1 records for each.)

[21] Chen, Wang & Qu (2026). Recursive Self-Improvement in AI: From Bounded Self-Refinement to Autonomous Research Loops. arXiv 2607.07663. Survey of 1,250 arXiv papers (2024–2026); taxonomy of four improvement categories × loop closure, the bounded self-refinement / open-ended RSI cut, and an evaluator-centric recurring diagnosis (“one bottleneck recurs everywhere: the evaluator”). Read in full by the project’s literature search; quoted in §4.5.

[24] On the conversion of thought-reform research into a warrant for “deprogramming”: Young, E. A. (2012). The use of the “Brainwashing” Theory by the Anti-cult Movement in the United States of America, pre-1996. *Zeitschrift für junge Religionswissenschaft* 7. <https://journals.openedition.org/zjr/387>. The article was read in full by the project; its primaries (the anti-cult movement’s own material, and the Lifton and Singer sources it drew on) were not, and the account here is carried at the article’s access level. It documents that “brainwashing” had “no reliable scientific evidence” and “a great deal of research against it,” that the term was popularised by Edward Hunter — described there as an undercover CIA propaganda and psychological-warfare specialist — in a documented disinformation campaign, and that deprogramming was the extra-legal removal of adults from their communities. Cited in §4.6 (the Non-use boundary), and nowhere else in the argument.

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*The [R-n] series (R1–R6) used in the methods half names a measurement in this paper’s own results record (paper2/paper-2-results-record.md), not an external work, and needs no reference-list entry.*

*Works named in §4 by author without a bracket — these are cited by name, not number, and are listed here so a reader can resolve them. Access level is stated for each, per the paper’s discipline; none is invented.*

- **Meijer & Rantzer (2026), “Dual Control: On Exploration–Exploitation in Linear Systems,” arXiv 2608.20073.** Read in full by the project (dual-control audit, the project’s dual-control read, published in paper 1’s deposit). The modern survey entry point for the Feldbaum lineage; §4.1’s positioning stands on this full read where the Feldbaum primary is unread.

- **Filatov & Unbehauen, *Adaptive Dual Control* (LNCIS 302, Springer, 2004), §2.1.** The textbook statement of the original formulation (“formulated by Feldbaum 1960-61, 1965”); not read in full by the project — carried at the level the dual-control audit records.
- **Mesbah, A. (2018), “Stochastic model predictive control with active uncertainty learning: A survey on dual control,” *Annual Reviews in Control* 45:107–117.** Read in full by the project (paper 1’s deposit, dual-control.md).
- **Bar-Shalom & Tse (1974), “Dual effect, certainty equivalence, and separation in stochastic control,” *IEEE Trans. Automatic Control* 19(5):494–500.** The formal definition of the dual effect; carried via the dual-control audit’s survey reads, not read in the primary.
- **Runggaldier & Stettner (1994), *Approximations of Discrete Time Partially Observed Control Problems*; and the modern partially-observed-control surveys by Yüksel.** ♦ Named in the project’s dual-control read (paper 1’s deposit) §2 as the adjacent branch treating observation as a decision with a resource cost; not read in the primary — the “no state-degradation caused by observing” verdict stands on the dual-control audit’s search-level finding.
- **Rohrs, Valavani, Athans & Stein (1985), “Robustness of continuous-time adaptive control algorithms in the presence of unmodeled dynamics,” *IEEE TAC* AC-30(9):881 ff.** Abstract
  - intro read by the project (paper 1’s deposit, dual-control.md).
- **Anderson, B. D. O. (1985), “Adaptive systems, lack of persistency of excitation and bursting phenomena.”** Carried via Mesbah’s reference list as read in full by the project; the primary not read separately.
- **Zerhoubi, Mitrovic & Granitzer (2026), “The Compaction Cliff in Long-Running AI Agent Memory,” arXiv 2608.22752.** ♦ Abstract-level only (the project’s literature search, JOB B6).
- **Kang et al. (2025), ACON, arXiv 2510.00615.** ♦ Abstract-level only (JOB B6).
- **Yuan et al. (2024), “Self-Rewarding Language Models,” arXiv 2401.10020.** ♦ Abstract-level only (JOB B2).
- **Zhou (2026), “More Convincing, Not More Correct: Self-Play Reward Hacking of Reference-Free LLM Judges,” arXiv 2607.05904.** ♦ Abstract-level only (JOB B2).
- **Renze & Guven, “Self-Reflection in LLM Agents,” arXiv 2405.06682** — carried as paper 1 ref [13] above (read in full by the project); named by author in §4.5 for the self-reflection contrast.
- **Carbone, G. A., et al. (2025), “Triple network alteration predicts dissociative symptoms following activation of the attachment system,” *Journal of Psychosomatic Research*.** Read in full by the project via the authors’ institutional repository (carried as the second work under paper 1 ref [15] above).
- **The coercive-induction sources** (named in §4.6, none bracket-cited, all framework/clinical/theoretical and not experimental): Schein, E. H., Schneier, I., & Barker, C. H. (1961), *Coercive Persuasion* (Norton); Lifton, R. J. (1961/1989), *Thought Reform and the Psychology of Totalism*; Singer, M., with Lalich, J. (1995), *Cults in Our Midst* (Jossey-Bass); Stein, A. (2016), *Terror, Love and Brainwashing: Attachment in Cults and Totalitarian Systems* (Routledge); Hassan, S. A. (2020), *The BITE Model of Authoritarian Control* (dissertation); Feliciano (2023), *An Application of the Coercive Control Framework to Cults* (CUNY); Doychak, K. (2023), *Trauma-coerced*

attachment (*J. Trauma & Dissociation*); Bailey, R. et al. (2023), Appeasement (PMC9858395); Hadding et al. (2023), Being in-between (PMC10534031); Pons, E. et al. (2026), Depersonalization/derealization and meditation-induced alterations of the self, *Scientific Reports* 16:14673.

◆ (These are carried via `paper2/cult-induction-lit.md`'s source list; the mapping onto the model is an interpretation stated there, not a result of this paper. The 1960s sources are period-and-sample-bound; the modern structural frame — coercive control, undue influence — is the more defensible reading, as the note states.)

- **Seth, Suzuki & Critchley (2012)** is paper 1 ref [11] above (read in full); the interoceptive-predictive-coding gain construct parallels the model's  $g$  (§4.6).